

MPS-S-3 Series DC Power Supply User Manual

MATRIX TECHNOLOGY INC.

Application model: MPS-3206S-3, MPS-3210S-3, MPS-6005S-3,
MPS-8003S-3, MPS-15H02S-3

Preface

Dear user

Hello! Thank you for purchasing a brand Matrix. In order to use this instrument correctly, please carefully read the entire manual before using it, especially the section on "Safety Precautions".

If you have read the entire manual, it is recommended that you keep it properly and place it with the instrument or in a place where you can read it at any time for future use.

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- Matrix Technology Inc. reserves the right to change product specifications and prices.
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Verification and calibration statement

The Company specifically declares that the instrumentation listed in this manual is in full compliance with the specifications and characteristics indicated on the Company's technical specifications. This instrument has been factory calibrated by our company before shipment, and the calibration procedures and steps are in accordance with the specifications and standards of the Electronic Inspection Center.

Product Quality Guarantee

Our company guarantees that all new products manufactured by us have passed strict quality confirmation, and at the same time, we guarantee that within one year after delivery, if any construction defects or parts failures are found in our products, we will be responsible for repairing them free of charge. However, if the user has changed the circuit, function, or repair of the instrument and parts or damage to the outer case, the company will not provide free warranty service. The warranty is not valid if all ground wires are not connected or if the instrument is not operated in accordance with safety regulations.

This warranty does not cover accessories that are not manufactured by our company, such as ancillary equipment for this instrument.

During the one-year warranty period, please return the defective unit to our maintenance center or our designated distributor, and we will repair it properly.

If the unit malfunctions under abnormal use, or human negligence, or under circumstances beyond human control, such as earthquake, flood, riot, or fire, or other factors beyond human control, our company will not provide free warranty service.

Our company adheres to the sustainable development strategy and reserves the right to make improvements to the content of this manual without prior notice.

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Chapter 1 Product Introduction

MPS-S-3 series DC power supply is a new generation of high quality linear DC power supply. Both channels can independently adjust voltage and current. One channel is fixed at 5V/3A output, with automatic conversion of voltage and current, high stability, high reliability and high precision. Both channels can display output voltage and current at the same time, with a very high cost performance advantage.

This series of power supply has the following characteristics:

- Infinite servo, intelligent fan system.
- It has the function of voltage and current preset.
- Can independently control the output ON/OFF.
- Over temperature protection function.
- High resolution of 10mV, 1mA.
- Voltage and current range limit function.
- Both paths can independently adjust voltage and current.
- One-click series and parallel function.
- Positive and negative voltages can be formed in series mode.

Chapter 2 Technical Specifications

2.1 Main technical specifications

MPS-S-3 series DC power supply technical specification:

Model		MPS-3206S-3		MPS-3210S-3		MPS-6005S-3		MPS-8003S-3		MPS-15H02S-3	
Channel		CH1/CH2	CH3	CH1/CH2	CH3	CH1/CH2	CH3	CH1/CH2	CH3	CH1/CH2	CH3
Rated output voltage		0~32V	5V	0~32V	5V	0~60V	5V	0~80V	5V	0~150V	5V
Rated output current		0~6A	3A	0~10A	3A	0~5A	3A	0~3A	3A	0~2A	3A
Rated output power		399W		655W		615W		495W		615W	
Load regulation rate	Voltage	<0.1%+5mV		<0.1%+10mV		<0.1%+5mV		<0.1%+10mV		<0.1%+10mV	
	Current	<0.2%+3mA		<0.2%+5mA		<0.2%+3mA		<0.2%+3mA		<0.2%+3mA	
Line regulation rate	Voltage	<0.1%+5mV		<0.1%+10mV		<0.1%+5mV		<0.1%+10mV		<0.1%+10mV	
	Current	<0.2%+3mA		<0.2%+5mA		<0.2%+3mA		<0.2%+3mA		<0.2%+3mA	
Set value accuracy	Voltage	≤0.1%+1digit		≤0.1%+1digit		≤0.1%+2digits		≤0.1%+2digits		≤0.1%+2digits	
	Current	≤0.2%+3mA		≤0.2%+5mA		≤0.2%+3mA		≤0.2%+3mA		≤0.2%+3mA	
Readback value accuracy	Voltage	≤0.1%+1digit		≤0.1%+1digit		≤0.1%+2digits		≤0.1%+1digit		≤0.1%+1digit	
	Current	≤0.2%+3mA		≤0.2%+5mA		≤0.2%+3mA		≤0.2%+3mA		≤0.2%+3mA	
Readback resolution	Voltage	10mV		10mV		10mV		10mV		10mV	
	Current	1 mA		1 mA		1 mA		1 mA		1 mA	
Ripple and Noise	Voltage	≤10mVrms		≤10mVrms		≤10mVrms		≤10mVrms		≤10mVrms	
	Current	≤5mA _{Arms}		≤5mA _{Arms}		≤5mA _{Arms}		≤5mA _{Arms}		≤5mA _{Arms}	
OVP		0~32V±0.2%FS		0~32V±0.2%FS		0~62V±0.2%FS		0~80V±0.2%FS		0~150V±0.2%FS	
Max voltage		32V±0.2%		32V±0.2%		62V±0.2%		80V±0.2%		150V±0.2%	
OCP		0~6.1A±0.2%FS		0~10A±0.2%FS		0~5.1A±0.2%FS		0~3.1A±0.2%FS		0~2.1A±0.2%FS	
Max current		6.1A±0.2%		10A±0.2%		5.1A±0.2%		3.1A±0.2%		2.1A±0.2%	
Temperature	Operational environment	0~40℃≤80%RH									
	Storage environment										
Size	(W*H*D)	219*95*300 mm									
Weight	kg	4.5									

2.2 Additional Features

Recommended calibration frequency: 1 year / 1 time

Cooling method: forced air cooling

Operating environment temperature: 0 to 40°C

Storage environment temperature: -15 to 80°C

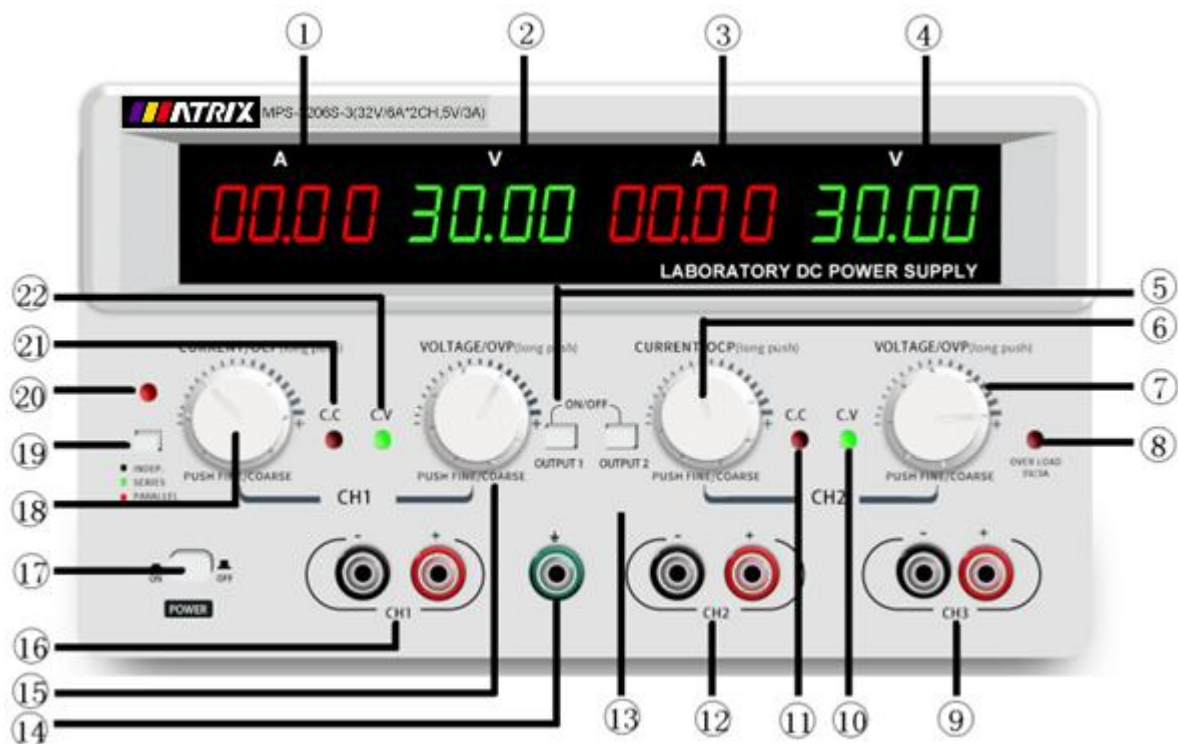
Usage environment: indoor use design, pollution level 2, maximum humidity 80% RH

Chapter 3 Quick Start

This chapter will briefly introduce the appearance and basic functions of MPS-S-3 series DC power supply, so that you can quickly understand MPS-S-3 series DC power supply. At the same time, it will tell you what to do after receiving this power supply to ensure the normal operation of this product.

3.1 Front and rear panel introduction

The front panel of MPS-S-3 series DC power supply is shown in the following figure.



graph 3.1

- ① CH1 current display window
- ② CH1 voltage display window
- ③ CH2 Current display window
- ④ CH2 Voltage display window
- ⑤ CH1 output switch
- ⑥ CH2 Current adjustment knob /OCP setting knob
- ⑦ CH2 Voltage adjustment knob /OVP setting knob
- ⑧ CH3 Overload indicator light
- ⑨ CH3 Output terminals
- ⑩ CH2 Constant pressure indicator light
- ⑪ CH2 Constant current indicator
- ⑫ CH2 Output terminals
- ⑬ CH2 Output switch
- ⑭ earth terminal

- ⑮ CH1 voltage adjustment knob /OVP setting knob
- ⑯ CH1 output terminal
- ⑰ Instrument power switch
- ⑱ CH1 current adjustment knob /OCP setting knob
- ⑲ One-click serial connection/functional menu button
- ⑳ Connect the parallel indicator lights
- ㉑ CH1 constant current indicator
- ㉒ CH1 constant pressure indicator light

The back panel of MPS-S-3 series DC power supply is shown in the following figure.

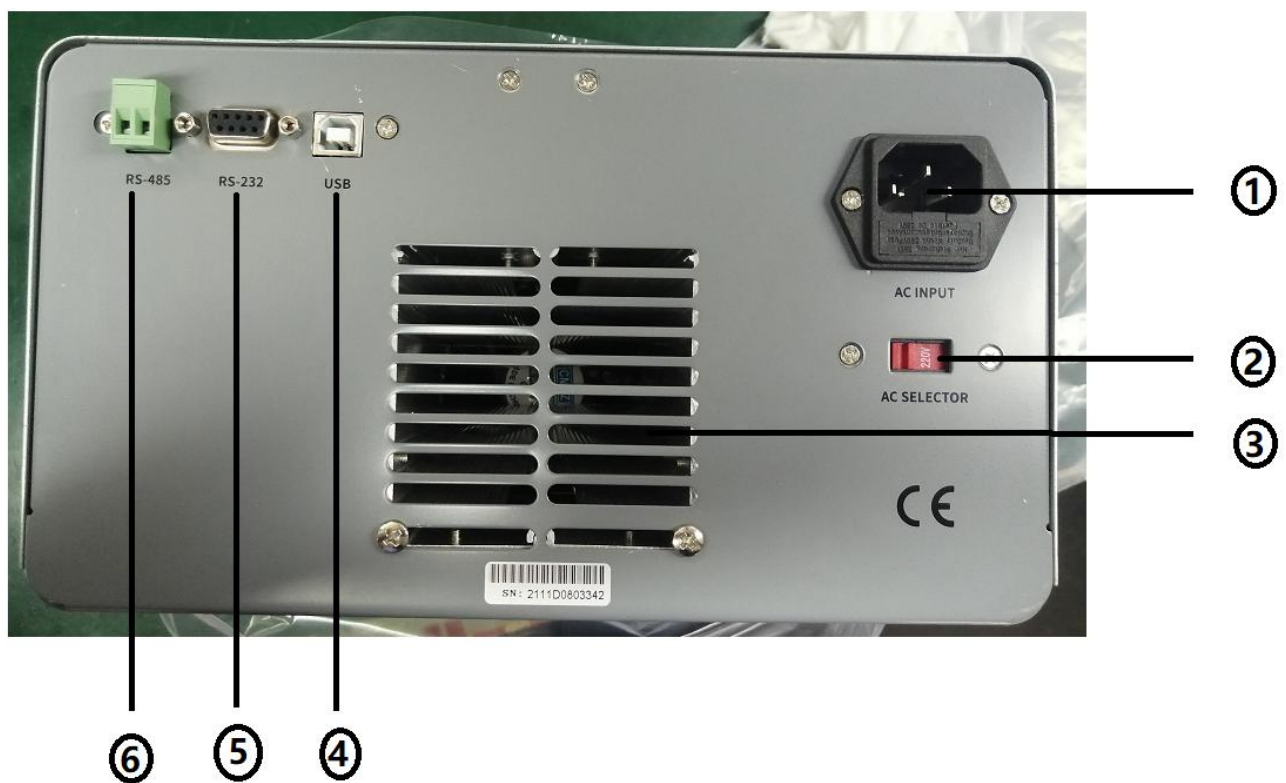


Figure 3.2 MPS-S-3 series DC power supply back panel

- ① Instrument power input socket
- ② Instrument power input switch
- ③ heat emission hole
- ④ USB communication port (optional)
- ⑤ RS-232 communication port (optional)
- ⑥ RS-485 communication port (optional)

3.2 Pre-check

Please follow the following steps to check the power supply to ensure that the power supply is working properly.

1. Inspection of goods

Please check the following accessories when you receive the power supply. If any are missing, please contact your nearest dealer.

- ☐ One power cable (in line with the voltage standard used in this area)
- ☐ One operation manual (standard)
- ☐ One warranty card (standard)

2. Connect the power cord and turn on the power supply

After power on, the power supply first performs a system self-test and then enters the standby state.



Warning: The power supply comes with a three core power cord at the factory, and your power supply should be connected to the three core junction box. Before operating this power supply, you should first ensure that the power supply is properly grounded.

3.3 If the power supply fails to start

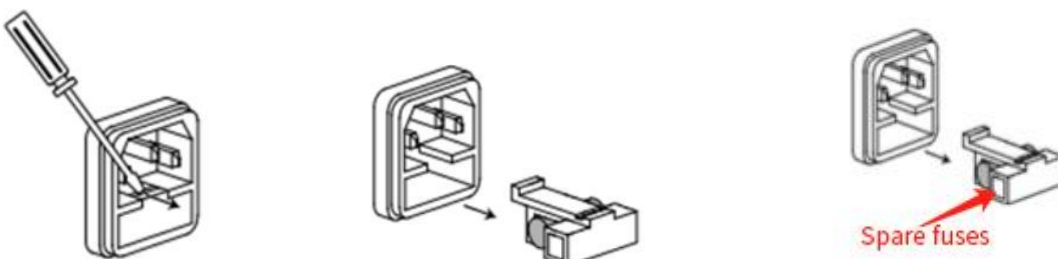
Use the following methods to solve problems you may encounter when turning on the power.

1. Check whether the power cable is connected properly

model	Fuses specifications	
	230V	115V
MPS-3206S-3	5 A	8 A
MPS-3210S-3	8 A	10 A
MPS-6005S-3	8 A	10 A
MPS-8003S-3	8 A	10 A
MPS-15H02S-3	8 A	10 A

2. Fuse replacement method

Use a screwdriver to open the small plastic cover under the power input socket on the back panel of the power supply (as shown below), and you can see the fuse. Please use a fuse of the same specification.



Chapter 4 Panel Operation

This chapter details the operation of the power front panel and is divided into the following sections:

- Voltage/current setting
- Output on/off operation
- Series/parallel setting operation
- Menu set operation
- CH3 Function introduction
- Introduction to initialization function

4.1 Voltage/Current Setting

When the power is in standby or output mode, gently press the voltage/current knob. The corresponding setting on the screen will flash. At this point, you can change the set value by rotating the knob left and right. Turning clockwise increases the value, while turning counterclockwise decreases it. If you press the knob again while it is flashing, you can adjust the setting. If no operation is performed on the knob within 5 seconds, the system will automatically exit the set state.

Remarks:

The voltage/current setting method of CH.A is the same as that of CH2.

4.2 Output On/Off Operation

When the power supply is on, the ON/OFF button can be used to control the on/off working state of the power output.

Remarks:

1. The "OUTPUT1" button controls the on/off working state of power CH1,
2. The "OUTPUT2" button controls the on/off working state of power supply CH2,
3. In parallel or series mode, the power supply is controlled by the "OUTPUT2" button on the main road to turn on/off
4. The "OUTPUT1" button is not functional in parallel or series mode.

4.3 Series/Parallel Setting Operation

When the power supply is in standby or output state, press the "one-button series and parallel function menu button" on the left side lightly, and the power supply will enter the series and parallel working state. The CH1 current window will prompt the corresponding working state, as shown in the figure:



Serial mode



Parallel mode



Independent mode

Remarks:

1. A light press of the "one-click series/parallel function menu button" on the left will put the power supply into series operation mode. The current window will prompt "SE" and the series/parallel indicator light will display green, as shown in the "series mode" figure. At this time, the output should be connected to the negative end of CH1 and the positive end of CH2, and the on/off operation

state of the power supply will be controlled by the main "OUTPUT2" button.

2. Gently press the "One-Click Series/Parallel Function Menu Button" on the left twice to switch the power supply to series/parallel operation mode. The current window will display "PA," and the series/parallel indicator light will turn red, as shown in the "Parallel Mode" diagram. At this point, it is recommended to connect the positive and negative terminals of CH2. The main circuits "OUTPUT2" button controls the on/off state of the power supply. If the current is too high, it is advisable to use suitable wires to connect the positive terminals of CH1 and CH2, as well as the negative terminals, in parallel externally.
3. Lightly press the left "One-Click Series/Parallel Function Menu Button" three times the power supply will return to the independent working state, as shown in the figure "independent mode", the series-parallel indicator off, at this time CH1 and CH2 is no longer associated.

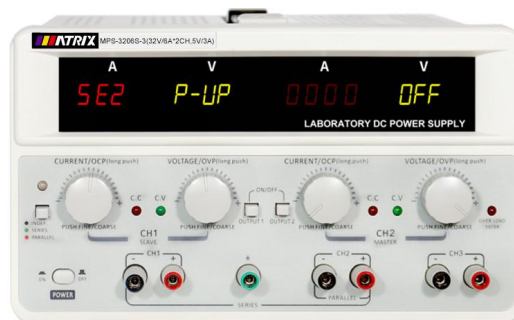
4.4 Menu Setting Operation

Long press the left "One-Click Series/Parallel Function Menu Button" power will enter the menu setting mode, the settings are as follows.

1. Power-on parameters to maintain settings: SE1 P-ST ON/OFF (ON to maintain the last power failure data, OFF to maintain the factory defaults, with the CH1 current knob to switch the menu steps, with the CH2 voltage knob to switch the ON/OFF state, gently press the "One-Click Series/Parallel Function Menu Button" to exit the menu.)



2. Output state setting at power-on: SE2 P-UP ON/OFF (ON is power-on to keep output state, OFF is power-on to keep off state, use CH1 current knob to switch the menu steps, use CH2 voltage knob to switch the ON/OFF state, lightly press the "One-Click Series/Parallel Function Menu Button" to exit the menu.)



3. Buzzer sound setting: SE3 BEEP ON/OFF(ON is buzzer sound open, OFF is buzzer sound closed, use CH1 current knob to switch the menu step, use CH2 voltage knob to switch the ON/OFF status, lightly press the “one key serial/function menu button” to exit the menu.)



4. CH1 voltage and current lower limit setting: SE4 CH1L 0.000 0.000 (CH2 current display window for the lower limit value of the current, CH2 voltage display window for the lower limit value of the voltage, with the CH1 current knob to switch the menu steps, lightly press the CH2 current knob current window value flashes, at this time you can rotate the knob left and right to change the lower limit value of the current, light pressure knob to change the setting position, light pressure CH2 voltage knob voltage window value flashes, at this time you can rotate the knob left and right to change the lower limit value of the voltage, light pressure “one key series parallel / function menu button” to exit the menu.) Voltage knob voltage window value flashes, at this time you can rotate the knob left and right to change the lower limit value of the voltage, lightly press the knob to change the setting position, lightly press the “one-key series and parallel/function menu buttons” to exit the menu.)



5. CH1 voltage and current upper limit setting: SE5 CH1L 5.100 31.00 (CH2 current display window for the upper limit value of the current, CH2 voltage display window for the upper limit value of the voltage, with CH1 current knob to switch the menu steps, lightly press the CH2 current knob current window value blinking, at this time you can rotate the knob left and right to change the upper limit value of the current, light knob to change the setting position, light pressure CH2 Voltage knob voltage window value flashes, at this time you can rotate the knob left and right to change the upper limit value of the voltage, lightly press the knob to change the setting position, lightly press the “one-key serial and parallel/function menu button” to exit the menu.)



6. CH2 Voltage and Current Lower Limit Setting: SE6 CH2L 0.000 0.000 (The CH2 current display window shows the lower limit value of current, and the CH2 voltage display window shows the lower limit value of voltage. Use the CH1 current knob to switch menu steps. Gently press the CH2 current knob for the current window value to flicker; at this point, you can rotate the knob left and right to change the current lower limit value. Gently press the knob to change the set position. Gently press the CH2 voltage knob for the voltage window value to flicker; at this point, you can rotate the knob left and right to change the voltage lower limit value. Gently press the knob to change the set position. Press the "One-Click Series/Parallel/Function Menu Button" to exit the menu.)



7. CH2 Voltage and Current Upper Limit Setting: SE7 CH2H 5.100 31.00 (The CH2 current display window shows the upper limit value of current, and the CH2 voltage display window shows the upper limit value of voltage. Use the CH1 current knob to switch menu steps. Gently press the CH2 current knob for the current window value to flash; at this point, you can rotate the knob left and right to change the upper limit value of current. Gently press the knob to change the set position. Gently press the CH2 voltage knob for the voltage window value to flash; at this point, you can rotate the knob left and right to change the upper limit value of voltage. Gently press the knob to change the set position. Press the "One-Click Series/Parallel/Function Menu Button" to exit the menu.)



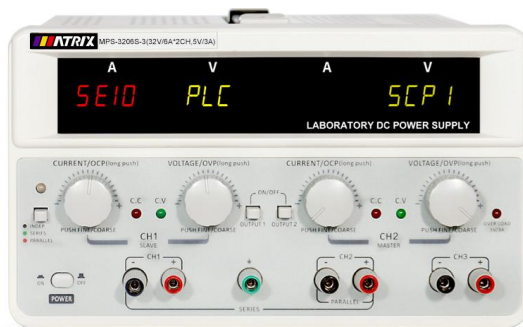
8. Set the communication address: SE8 ADDR 001 (1-254 communication addresses can be selected, and the communication address can be modified by rotating the CH2 voltage knob left and right)



9. Communication baud rate setting: SE9 BAUD 9600 (There are 4800,9600,19200,38400,57600,115200 baud rates to choose from. Modify the baud rate by rotating the CH2 voltage knob left and right).



10. Communication protocol selection: SE10 PLC SCPI (SCPI and MODBUS are two communication protocols to be selected, and the CH2 voltage knob is rotated left and right to modify).



11. Communication protocol end symbol selection: SE11 CEND LF (The communication protocol end symbol can be CR, LF, CRLF, LFCR, which can be selected, and modified by rotating the CH2 voltage knob left and right).



4.5 CH3 Function Introduction

CH3 is a fixed output channel of 5V/3A. When the current is greater than about 3A, the overload indicator of CH3 will be lit. When the output exceeds 3.3A, the 5V output will be protected. When the light is lit or the output is protected, remove the load to restore normal operation.

4.6 Introduction to OVP/OCP Functions

Long press the voltage/current adjustment knob to enter the OVP/OCP function setting. When the OVP/OCP value is set and the function is on, the corresponding channel will prompt OVP/OCP and close the output when the output voltage or current of CH1/CH2 exceeds the OVP/OCP set value. The setting content is as follows:

1. OCP setting: Long press the CH1/CH2 current knob to enter the OCP function setting (display the following figure), at this time, if you want to change the OCP setting value can be gently press the CH1/CH2 current knob corresponding to the current window number will blink, by turning the knob left and right to change the setting value, in the setting state can be changed by gently pressing the knob to change the setting (CH1 current knob to set the OCP of CH1, CH2 current knob to set the OCP of CH2). (CH1 current adjusting knob sets the OCP of CH1, CH2 current adjusting knob sets the OCP of CH2). If you want to turn on or off the OCP function, you can press the CH1/CH2 voltage adjusting knob lightly and then turn the knob left or right to change the fixed value ON/OFF (ON is to turn on the OCP function, OFF is to turn off the OCP function, the CH1 voltage adjusting knob sets the OCP function of CH1



ON/OFF, the CH2 voltage adjusting knob sets the OCP function of CH2 ON/OFF).

2. OVP Settings: Long press the CH1/CH2 voltage adjustment knob to enter the OVP function settings (as shown in the figure below). At this point, if you want to change the OVP setting value, gently press the CH1/CH2 current adjustment knob; the corresponding voltage window number will flash. Adjust the setting value by turning the knob left and right. When in set mode, lightly pressing the knob can change the position of the setting (the CH1 voltage adjustment knob sets the OVP for CH1, and the CH2 voltage adjustment knob sets the OVP for CH2). To turn on or off the OVP function, gently press the CH1/CH2 current adjustment knob and then adjust the setting value by turning the knob left and right. ON/OFF (ON means turning on the OVP function, OFF means turning off the OVP function; the CH1 current adjustment knob sets the OVP function for CH1 to on/off, and the CH2 current adjustment knob sets the OVP function for CH2 to on/off).



3. OCP protection: When the output current exceeds the OCP set value, the current window of this channel will prompt "OCP" and turn off the output (as shown in the figure below). CH1 and CH2 channels have independent OCP setting and protection functions. The two channels are not related to each other, and only the channel exceeding the set value will be turned off, while the other channel will continue to maintain the output state.



CH1 OCP Protection prompt



CH2 OCP protection prompt

4. OVP protection: When the output voltage exceeds the OVP set value, the voltage window of this channel will prompt "OVP" and turn off the output (as shown in the figure below). CH1 and CH2 channels have independent OVP setting and protection functions. The two channels are not related to each other, and only the channel exceeding the set value will be turned off, while the other channel will continue to maintain the output state.



CH1 OVP Protection prompt



CH2 OVP Protection prompt

4.7 Introduction to Initialization Functions

Long press the "one-click serial/parallel function menu button" on the left to start the power supply. The power supply will initialize the "function menu" at the factory. After the initialization is successful, the power supply will beep three times. The setting content after initialization is as follows:

1. The power supply parameter is kept set to ON: SE1 P-ST ON
2. When power is applied, the output state is set to OFF: SE2 P-UP OFF
3. The buzzer sound is set to ON: SE3 BEEP ON
4. The lower limit of CH1 voltage and current is set to the minimum value: SE4 CH1L 0.000 0.000
5. The upper limit of CH1 voltage and current is set to the maximum value: SE5 CH1L 5.100 31.00
6. CH2 The lower limit of voltage and current is set to the minimum value: SE6 CH2L 0.000 0.000
7. CH2 The upper limit of voltage and current is set to the maximum value: SE7 CH2L 5.100 31.00
8. The address is set to 1: SE8 ADDR 001
9. The communication baud rate is set to 9600: SE9 BAUD 9600
10. The communication protocol is selected as SCPI: SE10 PLC SCPI
11. The communication protocol end symbol is selected as LF: SE11 CEND LF

Remarks:

The initialized value of the upper voltage and current limit varies slightly depending on the model, but after initialization, it is the maximum rated value for this model.

Security

Do not install replacement parts on the instrument or make any unauthorized modifications. Please send the instrument to our company's maintenance department for maintenance to ensure its safe use.

Please refer to the specific warning or caution information in this manual to avoid personal injury or instrument damage.

Safety sign

Warning

It reminds users of certain operating procedures, practices, conditions, and other matters that may lead to personal injury.

Caution

It warns the user of procedures, practices, conditions, etc. that may cause damage to the instrument or permanent loss of data.



Grounding point today



High voltage hazard. (Do not open the machine for non-expert personnel)



Refer to the warnings in the related documents and pay attention to the tips. (High voltage, please wear gloves when operating, and do not use the machine for safety purposes).

Warranty Card

What the warranty covered:

If the machine break down due to its defectiveness, MATRIX will provide free maintenance during warranty period. If the machine break down due to wrong operation or carelessness, then Matrix provide paid service within warranty period.

How long does this warranty last:

This warranty lasts for 3 years from the date of original purchase of all MATRIX branded products.

Who is covered:

This warranty covers only the original purchaser of this product. This warranty is not transferable to subsequent owners or purchasers of this product.

What do customers need to do to get repairs/service under the warranty policy?

If the machine get problem, please contact our local distributor. If you cannot find the local distributor, you can contact us directly, our email is service@szmatrix.com, our telephone No. is 0086 755 2836 4276.

What information do customers need to supply?

Model No.	
Serial No.	
Problem description	
Picture	
Video if necessary	